

Chapter 6A

Endocrine Overview: Hormone Axes, Feedback Loops, Diabetes Screening, and Metabolic Emergencies

Medvale Internal Medicine | Part VI: Endocrinology and Metabolism

This chapter is an original study resource designed for clinical reasoning and future graph-RAG extraction. Uploaded source sections on diabetes, thyroid disease, adrenal disease, and pituitary/water-balance disorders were used as coverage signals; the explanatory prose, tables, diagrams, and cases are newly written.

Learning objectives

- Build a first-pass endocrine differential from the pattern of symptoms: hormone excess, hormone deficiency, receptor resistance, gland mass effect, or metabolic emergency.
- Interpret endocrine laboratory results by locating the abnormality in the axis: peripheral gland, pituitary, hypothalamus, ectopic source, medication effect, or binding-protein artifact.
- Recognize the numerical anchors for diabetes screening, thyroid testing, adrenal testing, pituitary mass evaluation, calcium/parathyroid physiology, and endocrine emergencies.
- Connect endocrine diagnoses to mechanisms, complications, treatments, and high-yield pitfalls in a way that can be converted into graph nodes and edges.

1. Why endocrine disease matters clinically

Endocrinology is the discipline of delayed cause and amplified effect. A small gland, a tiny peptide concentration, a transcription factor, or a receptor mutation can produce organ-system disease that looks cardiac, psychiatric, neurologic, renal, reproductive, infectious, or gastrointestinal. For a learner, endocrine disease becomes manageable when every problem is translated into three questions: what hormone signal is abnormal, where in the axis is the abnormality located, and what target-organ effect explains the patient's presentation?

Many endocrine presentations are outpatient and chronic, such as weight change, fatigue, menstrual irregularity, infertility, hypertension, dyslipidemia, osteoporosis, polyuria, or incidental nodules. Others are immediate emergencies: diabetic ketoacidosis, hyperosmolar hyperglycemic state, thyroid storm, myxedema coma, adrenal crisis, severe hypercalcemia, hypocalcemic tetany, pituitary apoplexy, and symptomatic hyponatremia. The same patient can move between chronic and acute states; for example, a patient with type 1 diabetes may be stable for years but develop DKA during infection, or a patient with chronic adrenal insufficiency may collapse during sepsis if stress-dose steroids are not provided.

Core endocrine reasoning loop	
1. Identify syndrome	Excess vs deficiency vs mass effect vs metabolic emergency
2. Confirm hormone abnormality	Use the screening test that best reflects physiology, not necessarily the most familiar lab
3. Localize the source	Compare tropic hormone to target hormone: TSH/T4, ACTH/cortisol, LH/FSH/sex steroids, PTH/calcium
4. Explain complications	Map hormone action to organ injury: vascular, renal, bone, cardiac, neuropsychiatric, reproductive
5. Treat safely	Stabilize emergencies first; replace deficient hormone cautiously; block excess hormone and locate tumor/source

2. The language of endocrine physiology

Most endocrine hormones are either peptide hormones, steroid hormones, amine derivatives, or mineral ions acting through hormone-like pathways. Peptide hormones such as insulin, ACTH, TSH, LH, FSH, prolactin, growth hormone, and ADH generally act through membrane receptors and second messengers. Steroid hormones such as cortisol, aldosterone, estrogen, progesterone, and testosterone diffuse across cell membranes and alter gene transcription through intracellular receptors. Thyroid hormone is synthesized from tyrosine and iodine, circulates mostly bound to proteins, enters cells, and regulates transcription after conversion of T4 to the more active T3 in peripheral tissues.

The endocrine system also uses carrier proteins and reservoirs. Most circulating thyroid hormone is protein-bound, so total T4 and total T3 can change when thyroid-binding globulin changes, even when the free biologically active hormone is normal. Cortisol binds corticosteroid-binding globulin and albumin. Calcium is divided into ionized calcium, albumin-bound calcium, and complexed calcium; symptoms correlate best with ionized calcium. Vitamin D is stored and activated in steps across skin, liver, and kidney. These details create many exam traps: a lab can be abnormal because of binding, albumin, kidney disease, medication, acute illness, assay interference, or nonadherence rather than true gland disease.

Hormone class	Typical examples	Receptor/signaling logic	Clinical testing implication
Peptide/protein	Insulin, glucagon, ACTH, TSH, LH/FSH, prolactin, GH, ADH	Membrane receptors; rapid second-messenger effects; short half-life	Timing and pulsatility matter. Single random values may be misleading for GH or cortisol.
Steroid	Cortisol, aldosterone, estrogen, testosterone	Intracellular/nuclear receptors; gene transcription; protein-bound fraction	Measure axis context: ACTH with cortisol; renin with aldosterone; LH/FSH with sex steroids.
Amine/iodinated	T3, T4, catecholamines	Thyroid hormone is nuclear/transcriptional; catecholamines act via adrenergic receptors	Free T4 is often more interpretable than total T4; metanephrines are preferred for catecholamine-secreting tumors.
Mineral-regulatory	PTH, vitamin D, calcitonin, FGF23	Controls gut absorption, renal handling, and bone exchange of calcium/phosphate	Interpret calcium with albumin, phosphate, PTH, kidney function, and vitamin D status.

3. Feedback loops and axis localization

A classic endocrine axis has a hypothalamic releasing hormone, a pituitary tropic hormone, a peripheral gland hormone, and negative feedback from the peripheral hormone back to the pituitary and hypothalamus. Disease localization comes from whether the tropic hormone is appropriately high, appropriately low, or inappropriately normal. A low target hormone with a high tropic hormone suggests primary peripheral gland failure. A low target hormone with a low or inappropriately normal tropic hormone suggests central disease. A high target hormone with a suppressed tropic hormone usually suggests primary peripheral excess, exogenous hormone, or autonomous secretion. A high target hormone with a non-suppressed tropic hormone suggests pituitary or ectopic production, hormone resistance, or assay interference.

The pituitary is not just a hormone source; it is also a mass in a small space. Pituitary adenomas can cause hormone hypersecretion, hypopituitarism by compression of normal pituitary tissue, and mass effects such as headache or bitemporal hemianopsia from optic chiasm compression. A single lesion may therefore produce both excess and deficiency phenotypes: a macroadenoma secreting prolactin may also reduce gonadotropins, impair other axes, and cause visual field deficits.

Axis	Screening pattern	Primary gland disease	Central disease	Key trap
Thyroid	TSH first in most stable outpatients; add free T4 when abnormal or central disease suspected	Primary hypothyroidism: high TSH, low free T4. Primary hyperthyroidism: low TSH, high free T4/T3.	Central hypothyroidism: low/normal TSH with low free T4.	Normal TSH does not rule out central hypothyroidism in pituitary disease.
Adrenal cortisol	Morning cortisol, ACTH, cosyntropin testing; suppression tests for Cushing	Primary adrenal insufficiency: high ACTH, low cortisol, often hyperkalemia. Adrenal Cushing: high cortisol, low ACTH.	Secondary AI: low/normal ACTH and low cortisol. Cushing disease: ACTH-dependent cortisol excess.	Do not delay stress-dose steroids in adrenal crisis.
Aldosterone	Aldosterone-renin ratio in resistant HTN or hypokalemic HTN	Primary aldosteronism: high aldosterone, suppressed renin, HTN, hypokalemia, metabolic alkalosis.	Low renin states and medication effects complicate screening.	Potassium may be normal in primary aldosteronism.
Gonadal	LH/FSH with estradiol or testosterone	Primary gonadal failure: high LH/FSH and low sex steroids.	Hypogonadotropic hypogonadism: low/normal LH/FSH and low sex steroids.	Hyperprolactinemia suppresses GnRH and can mimic primary reproductive disease.
Water balance	Serum sodium/osmolality with urine osmolality and urine sodium	Nephrogenic DI: kidneys fail to respond to ADH; urine remains dilute after desmopressin.	Central DI: ADH deficiency; urine concentrates after desmopressin.	SIADH causes euvolemic hyponatremia without edema because natriuresis limits volume expansion.

4. How to interpret endocrine labs without getting trapped

Endocrine tests are powerful because they compress physiology into numbers, but they are also vulnerable to timing, binding proteins, kidney function, medications, acute illness, pregnancy, obesity, and assay artifacts. Before labeling a patient with endocrine disease, confirm that the test was obtained under interpretable conditions. A random cortisol in the afternoon is less useful than an 8 AM cortisol; a random growth hormone is less useful than IGF-1 because GH is pulsatile; total thyroid hormone values can move with thyroid-binding globulin; serum calcium must be interpreted with albumin or ionized calcium; A1c can mislead when RBC lifespan is abnormal; and urine sodium can be distorted by diuretics.

The most useful endocrine labs often come in pairs. TSH without free T4 may miss central hypothyroidism. Cortisol without ACTH may not localize adrenal disease. Aldosterone without renin cannot distinguish autonomous mineralocorticoid production from secondary RAAS activation. Calcium without PTH cannot separate parathyroid from nonparathyroid hypercalcemia. Prolactin without pregnancy testing and TSH may lead to unnecessary imaging. Serum sodium without serum osmolality and urine osmolality may confuse true hypotonic hyponatremia with pseudohyponatremia or translocational hyponatremia.

Lab problem	Do not overinterpret until you check...	Common explanation of false direction	Preferred next step
Abnormal A1c	Anemia, hemoglobinopathy, transfusion, CKD, pregnancy, rapid glycemic change	A1c reflects glycation over RBC lifespan; altered RBC turnover can falsely lower or raise it.	Use plasma glucose/CGM/fructosamine context when needed.
Low TSH	Free T4/T3, pregnancy, biotin use, acute illness, exogenous thyroid hormone	Nonthyroidal illness and medications can suppress TSH; T3 toxicosis may have normal free T4.	Repeat and add free T4/T3 if clinically stable; urgent evaluation if thyroid storm signs.
High prolactin	Pregnancy, hypothyroidism, renal failure, dopamine-blocking drugs, macroprolactin	Loss of dopamine inhibition or reduced clearance can elevate prolactin without prolactinoma.	Pregnancy test, TSH, medication review, then pituitary MRI if persistent/unexplained.
High cortisol	Exogenous steroids, depression/alcohol, obesity, acute illness, estrogen effect on binding globulin	Stress states can mimic Cushing biochemically; binding proteins affect total cortisol.	Use validated screening tests and confirm abnormal results before localization.
High calcium	Albumin, ionized calcium, PTH, malignancy clues, vitamin D, medications	Low albumin lowers total calcium without lowering ionized calcium.	Confirm true hypercalcemia; then separate PTH-mediated from non-PTH-mediated.
Hyponatremia	Serum osmolality, urine osmolality, urine sodium, glucose, volume status, thyroid/adrenal function	Hyperglycemia shifts water into ECF; SIADH requires exclusion of adrenal and thyroid disease.	Classify by tonicity, then urine osmolality, then volume/urine sodium.

5. The high-yield endocrine axes

Thyroid axis overview

The thyroid axis begins with hypothalamic TRH, pituitary TSH, thyroidal T4/T3 production, and peripheral conversion of T4 to T3. In most stable patients, TSH is the best screening test because the pituitary amplifies small changes in circulating thyroid hormone. Hyperthyroidism typically produces low TSH with high free T4 or T3, while primary hypothyroidism produces high TSH with low free T4. Graves disease adds autoimmune stimulation of the TSH receptor and can cause diffuse goiter, ophthalmopathy, pretibial myxedema, and thyroid bruit. Toxic multinodular goiter and toxic adenoma produce autonomous hormone with nodular uptake. Thyroiditis causes leakage of preformed hormone and often has low radioactive iodine uptake because new hormone synthesis is not increased.

Thyroid nodules are approached by TSH and ultrasound risk pattern, not by palpation alone. A low TSH raises the possibility of an autonomous hyperfunctioning nodule, which is less likely malignant. A normal or high TSH generally moves evaluation toward ultrasound risk stratification and fine-needle aspiration when size and sonographic features meet criteria. Thyroid cancer subtypes have different behavior: papillary is most common and lymphatic-spreading; follicular spreads hematogenously and requires capsular/vascular invasion for diagnosis; medullary arises from parafollicular C cells and produces calcitonin; anaplastic is aggressive and often presents with compressive invasion.

Adrenal axis overview

The adrenal cortex has three zones: zona glomerulosa makes aldosterone, zona fasciculata makes cortisol, and zona reticularis makes adrenal androgens. Aldosterone is primarily regulated by angiotensin II and potassium, not ACTH. Cortisol is

regulated by hypothalamic CRH and pituitary ACTH, and excess cortisol suppresses CRH/ACTH unless the source is ACTH-dependent. Adrenal medulla chromaffin cells produce catecholamines and are evaluated differently from cortical disease. This split explains why primary adrenal insufficiency causes both cortisol deficiency and mineralocorticoid deficiency, while secondary adrenal insufficiency usually preserves aldosterone because RAAS remains intact.

Cushing syndrome refers to chronic glucocorticoid excess. The most common overall cause is exogenous steroid exposure, but endogenous cases are localized using ACTH: low ACTH suggests adrenal cortisol production, while high or inappropriately normal ACTH suggests pituitary Cushing disease or ectopic ACTH. Primary aldosteronism should be suspected in resistant hypertension, hypertension with hypokalemia, adrenal incidentaloma with hypertension, or early-onset hypertension/stroke family patterns. Pheochromocytoma and paraganglioma cause episodic or sustained catecholamine excess; headache, sweating, palpitations, tachycardia, and paroxysmal hypertension are classic clues. Treatment sequencing is critical: alpha blockade before beta blockade prevents unopposed alpha-mediated vasoconstriction.

Pituitary and water-balance overview

Pituitary disease produces syndromes of excess secretion, deficiency, and local compression. Prolactinomas are treated medically with dopamine agonists in most cases. Acromegaly results from GH excess after epiphyseal closure and is screened with IGF-1; failure of GH suppression during oral glucose testing supports the diagnosis. Hypopituitarism produces loss of target hormones; gonadotropins and GH are often lost before TSH and ACTH, but ACTH deficiency is the most immediately dangerous because of adrenal crisis risk.

ADH disorders are framed by serum sodium and urine concentration. Diabetes insipidus causes polyuria with inappropriately dilute urine; central DI reflects ADH deficiency and responds to desmopressin, while nephrogenic DI reflects renal resistance and does not respond adequately. SIADH causes hypotonic euvolemic hyponatremia with inappropriately concentrated urine and continued urine sodium excretion. In SIADH, edema is absent because volume expansion triggers natriuresis, preventing large sustained extracellular fluid expansion.

Syndrome	Core hormone problem	Fast clinical pattern	Key test pattern	First dangerous thing to address
Graves hyperthyroidism	TSH receptor stimulation	Weight loss, tremor, heat intolerance, tachycardia, diffuse goiter; eye findings may occur	Low TSH, high free T4/T3, positive TRAb/TSI, diffuse uptake	Atrial fibrillation, thyroid storm, severe adrenergic symptoms
Primary hypothyroidism	Thyroid hormone deficiency from thyroid gland failure	Fatigue, cold intolerance, constipation, bradycardia, delayed reflexes	High TSH, low free T4	Myxedema coma if hypothermic/altered/hypoventilating
Adrenal crisis	Acute cortisol deficiency +/- mineralocorticoid deficiency	Hypotension, vomiting, abdominal pain, weakness, hyponatremia, hyperkalemia in primary AI	Low cortisol; ACTH high in primary, low/normal in secondary	Give hydrocortisone and IV fluids without waiting if unstable
Pheochromocytoma	Catecholamine excess	Headache, sweating, palpitations, tachycardia, episodic severe HTN	Plasma free or urine fractionated metanephrines elevated	Control BP with alpha blockade before beta blockade
Prolactinoma	Prolactin excess suppresses GnRH	Amenorrhea, infertility, galactorrhea; men often present later with mass effect	High prolactin; evaluate pregnancy, TSH, meds; MRI if persistent	Visual compromise from macroadenoma
DI	Absent ADH effect	Large-volume dilute urine, thirst, hypernatremia if unable to drink	High/normal serum osmolality with dilute urine; central responds to desmopressin	Free-water deficit and hypernatremia correction safety
SIADH	Excess ADH effect despite low osmolality	Euvolemic hyponatremia, neurologic symptoms if acute/severe	Low serum Osm, urine Osm not maximally dilute, urine Na often >30	Seizures/coma; avoid overly rapid sodium correction

6. Diabetes and metabolic screening framework

Diabetes is both an endocrine disease and a vascular disease. Hyperglycemia is the measurable abnormality, but long-term injury is mediated through endothelial dysfunction, oxidative stress, inflammation, advanced glycation end products, dyslipidemia, hypertension, kidney injury, retinal injury, neuropathy, and impaired immunity. A diabetes visit therefore cannot stop at glucose; it must also address blood pressure, lipids, kidney protection, retinal screening, neuropathy/foot examination, vaccinations, smoking cessation, obesity, psychosocial barriers, and medication safety.

Diagnostic thresholds are numerical anchors. Diabetes can be diagnosed with A1c $\geq 6.5\%$, fasting plasma glucose ≥ 126 mg/dL, 2-hour plasma glucose ≥ 200 mg/dL after a 75-g oral glucose tolerance test, or random plasma glucose ≥ 200 mg/dL with classic hyperglycemic symptoms. In the absence of unequivocal hyperglycemia, abnormal screening tests should be

confirmed. Prediabetes is commonly defined by A1c 5.7%-6.4%, fasting plasma glucose 100-125 mg/dL, or 2-hour glucose 140-199 mg/dL. These thresholds should be interpreted in clinical context, especially when A1c reliability is reduced.

Category	A1c	Fasting plasma glucose	2-hour OGTT	Random glucose
Normal	<5.7%	<100 mg/dL	<140 mg/dL	Not used alone
Prediabetes / impaired glucose regulation	5.7%-6.4%	100-125 mg/dL	140-199 mg/dL	Not used alone
Diabetes	>=6.5%	>=126 mg/dL	>=200 mg/dL	>=200 mg/dL with classic symptoms or crisis

Distinguishing type 1 from type 2 diabetes is sometimes easy and sometimes not. Type 1 diabetes reflects autoimmune beta-cell destruction and absolute insulin deficiency; DKA may be the first presentation. Type 2 diabetes reflects insulin resistance with progressive beta-cell dysfunction; obesity, acanthosis nigricans, hypertension, hypertriglyceridemia, fatty liver disease, and family history support the pattern. Adults can develop autoimmune diabetes, and patients with type 2 diabetes can develop ketosis under stress, so classification should remain evidence-based rather than purely age-based.

Medication selection in type 2 diabetes increasingly depends on comorbidities. Metformin remains a common foundational agent when tolerated and not contraindicated. SGLT2 inhibitors and GLP-1 receptor agonists are important when cardiovascular disease, heart failure, chronic kidney disease, obesity, and atherosclerotic risk shape therapy. Insulin is required in type 1 diabetes, pregnancy-specific situations, catabolic severe hyperglycemia, DKA, and many hospitalized or advanced type 2 diabetes contexts. Always pair intensification with hypoglycemia risk assessment, kidney function review, cost/access barriers, and patient education.

Drug/class	Main physiologic idea	High-yield benefits	High-yield cautions
Insulin	Replaces or supplements endogenous insulin; suppresses ketogenesis	Essential in type 1 DM, DKA/HHS transition, severe catabolic hyperglycemia	Hypoglycemia, weight gain; dose changes with illness, renal function, nutrition
Metformin	Lowers hepatic glucose output and improves insulin sensitivity	Weight-neutral/weight-lowering tendency; inexpensive; common first-line backbone	GI effects; avoid/hold in selected severe renal dysfunction or acute illness/contrast-risk scenarios
GLP-1 receptor agonists	Enhance glucose-dependent insulin, suppress glucagon, slow gastric emptying, increase satiety	Weight loss; ASCVD benefit for selected agents	GI effects; avoid in certain medullary thyroid cancer/MEN2 risk contexts depending on agent labeling
SGLT2 inhibitors	Increase urinary glucose excretion through proximal tubule SGLT2 blockade	Heart failure and CKD benefits for selected patients; modest weight/BP reduction	Genital infections, volume depletion, euglycemic ketoacidosis risk, perioperative holds
Sulfonylureas	Close beta-cell KATP channels to increase insulin release	Effective and inexpensive	Hypoglycemia and weight gain; caution in older adults/renal disease
DPP-4 inhibitors	Increase endogenous incretin effect	Weight-neutral, low hypoglycemia risk	Modest A1c effect; dose adjustment for many agents in CKD

7. Endocrine emergencies: recognize, stabilize, then localize

Endocrine emergencies are dangerous because they often masquerade as nonspecific shock, delirium, arrhythmia, infection, renal failure, or intoxication. The safest approach is to stabilize physiology first and refine diagnosis second. In suspected adrenal crisis, give stress-dose hydrocortisone and isotonic fluids immediately if the patient is unstable; diagnostic samples can be drawn first only if this does not delay treatment. In DKA and HHS, restore intravascular volume, monitor potassium, close the anion gap or resolve hyperosmolality safely, and search for triggers such as infection, myocardial infarction, stroke, medication omission, or steroid use. In thyroid storm, treat adrenergic excess, block new hormone synthesis, block hormone release, reduce peripheral T4-to-T3 conversion, and treat the precipitating illness.

Emergency	Hallmark physiology	Clues	Initial treatment priorities	Pitfall
DKA	Insulin deficiency with ketogenesis, anion gap metabolic acidosis, osmotic diuresis	Glucose often >250, ketones, low bicarbonate, low pH, high anion gap	IV fluids, potassium assessment/repletion, insulin after K is safe, trigger search	Serum K may be high while total body K is depleted.
HHS	Severe hyperglycemia and hyperosmolality with minimal ketosis	Very high glucose, dehydration, mental status change, older type 2 pattern	Aggressive fluids, electrolyte monitoring, insulin after volume and K assessment	Rapid osmolar shifts can worsen cerebral injury.

Emergency	Hallmark physiology	Clues	Initial treatment priorities	Pitfall
Thyroid storm	Extreme thyrotoxicosis plus systemic decompensation	Fever, tachyarrhythmia, agitation/delirium, GI symptoms, heart failure	Beta blocker, thionamide, iodine after thionamide, glucocorticoid, cooling/support	Giving iodine before thionamide can provide substrate for hormone synthesis.
Myxedema coma	Severe hypothyroidism with hypometabolism	Hypothermia, altered mental status, bradycardia, hypoventilation, hyponatremia	IV thyroid hormone, stress-dose steroids until adrenal insufficiency excluded, warming, ventilation support	Do not forget concurrent adrenal insufficiency risk.
Adrenal crisis	Cortisol deficiency +/- aldosterone deficiency causing shock	Hypotension, vomiting, abdominal pain, hyponatremia, hyperkalemia in primary AI	Hydrocortisone, isotonic saline with dextrose as needed, treat precipitant	Waiting for confirmatory testing can be fatal.
Severe hypercalcemia	High calcium causing renal vasoconstriction, dehydration, altered mental status, arrhythmia risk	Constipation, stones, confusion, shortened QT, dehydration	IV fluids, calcitonin for rapid bridge, antiresorptive therapy depending on cause/renal status	Treating without assessing volume and kidney function.
Symptomatic hyponatremia	Hypotonic cerebral edema	Seizure, coma, severe confusion, acute sodium drop	Hypertonic saline boluses for severe symptoms; monitor correction closely	Overcorrection risks osmotic demyelination.

8. Endocrine hypertension and electrolyte patterns

Endocrine causes of hypertension are high yield because they are often treatable and because the electrolyte pattern points to the mechanism. Primary aldosteronism causes sodium retention, potassium loss, hypertension, and metabolic alkalosis; renin is suppressed because volume expansion turns off RAAS. Cushing syndrome can cause hypertension through glucocorticoid effects, vascular sensitivity, weight gain, insulin resistance, and mineralocorticoid receptor activation at high cortisol levels. Pheochromocytoma causes episodic or sustained hypertension through catecholamine surges. Thyroid disease can alter blood pressure by changing cardiac output and vascular resistance: hyperthyroidism often produces systolic hypertension and wide pulse pressure; hypothyroidism may produce diastolic hypertension through increased systemic vascular resistance.

Hypertension clue	Think of...	Associated labs/features	Mechanism
Resistant HTN or HTN + hypokalemia	Primary aldosteronism	Low renin, high aldosterone-renin ratio, metabolic alkalosis; K can be normal	Autonomous aldosterone increases distal Na reabsorption and K/H secretion.
Paroxysmal HTN with headache/sweating/palpitations	Pheochromocytoma/paraganglioma	Elevated plasma free or urine fractionated metanephrines	Catecholamine excess stimulates alpha/beta adrenergic receptors.
HTN + central obesity + bruising + proximal weakness + purple striae	Cushing syndrome	Abnormal cortisol screening; ACTH helps localization	Cortisol increases vascular tone, glucose, catabolism, and mineralocorticoid activity.
HTN + tachycardia + tremor + weight loss	Hyperthyroidism	Low TSH with high free T4/T3	Increased beta-adrenergic tone and cardiac output.
HTN + episodic weakness + family/young onset	Rare mineralocorticoid excess syndromes	Low renin with variable aldosterone depending on syndrome	ENaC activation or steroid intermediates mimic mineralocorticoid excess.

9. Calcium, phosphate, bone, and endocrine metabolism

Calcium disorders connect endocrine physiology to renal function, malignancy, bone health, neuromuscular excitability, and arrhythmia risk. PTH raises serum calcium by increasing bone resorption, increasing renal calcium reabsorption, decreasing renal phosphate reabsorption, and stimulating renal 1-alpha hydroxylase to increase active vitamin D. Vitamin D increases intestinal absorption of calcium and phosphate. Calcitonin is less clinically central but opposes osteoclast activity. Kidney disease disrupts phosphate excretion and vitamin D activation, driving secondary hyperparathyroidism and renal osteodystrophy.

Hypercalcemia should be classified as PTH-mediated or non-PTH-mediated. If PTH is high or inappropriately normal, consider primary hyperparathyroidism, tertiary hyperparathyroidism, familial hypocalciuric hypercalcemia, or lithium effect. If PTH is suppressed, consider malignancy through PTHrP or osteolytic metastases, vitamin D excess, granulomatous disease, thyrotoxicosis, adrenal insufficiency, immobilization, medications, and milk-alkali syndrome. Hypocalcemia should be evaluated with magnesium, phosphate, PTH, vitamin D, albumin/ionized calcium, and renal function; severe symptoms include tetany, seizures, prolonged QT, laryngospasm, and arrhythmia.

Pattern	PTH	Phosphate	Likely categories	Clinical clue
Hypercalcemia with high/inappropriately normal PTH	High or non-suppressed	Often low in primary HPT; high in tertiary/CKD context	Primary hyperparathyroidism, tertiary hyperparathyroidism, lithium, FHH	Stones, bones, constipation, neuropsychiatric symptoms; often asymptomatic.
Hypercalcemia with low PTH	Suppressed	Variable	Malignancy, vitamin D excess/granulomatous disease, thyrotoxicosis, medications	Weight loss, cancer symptoms, high 1,25-OH ₂ D in granulomatous disease.
Hypocalcemia with high PTH	High	High in CKD; low in vitamin D deficiency	Vitamin D deficiency, CKD, pseudohypoparathyroidism	Tetany or bone pain; kidney labs distinguish mechanisms.
Hypocalcemia with low PTH	Low	High/normal	Hypoparathyroidism, severe hypomagnesemia	Post-thyroid surgery or autoimmune context.

10. Reproductive endocrine overview

Reproductive endocrine problems are best approached through the hypothalamic-pituitary-gonadal axis. GnRH drives LH and FSH release. LH stimulates Leydig cell testosterone production in men and theca cell androgen production/ovulation physiology in women. FSH supports spermatogenesis through Sertoli cells and follicular development/granulosa aromatase activity in women. Prolactin suppresses GnRH, which explains amenorrhea, infertility, decreased libido, and hypogonadism in hyperprolactinemia. Thyroid disease, adrenal disease, chronic illness, undernutrition, obesity, excessive exercise, pregnancy, medications, and pituitary tumors can all disrupt the axis.

Amenorrhea is never an endocrine puzzle before it is a pregnancy test. After pregnancy is excluded, TSH and prolactin are common early labs because thyroid disease and hyperprolactinemia are treatable and common mimics. Gonadotropin patterns then help separate ovarian failure from central hypogonadism. PCOS is suggested by oligo/anovulation, hyperandrogenism, and polycystic ovarian morphology after excluding mimics such as pregnancy, thyroid disease, hyperprolactinemia, nonclassic congenital adrenal hyperplasia, Cushing syndrome, and androgen-secreting tumors. Male hypogonadism is evaluated with morning testosterone confirmed on repeat testing and localized with LH/FSH.

11. Pattern-recognition maps for rapid differential diagnosis

Presenting pattern	Endocrine diagnoses to rapidly consider	First-pass tests
Weight loss + tachycardia + tremor	Hyperthyroidism, pheochromocytoma, uncontrolled diabetes, adrenal insufficiency, malignancy/infection	TSH/free T ₄ , glucose/A1c/ketones if ill, electrolytes, cortisol if hypotensive, metanephrines if episodic adrenergic spells
Weight gain + proximal weakness + bruising	Cushing syndrome, hypothyroidism, medication/steroid exposure, PCOS/insulin resistance	Medication history, late-night salivary cortisol/1-mg DST/24-h urine cortisol, TSH/free T ₄ , glucose/A1c
Polyuria + polydipsia	Diabetes mellitus, central DI, nephrogenic DI, primary polydipsia, hypercalcemia, hypokalemia	Serum glucose, sodium/osmolality, urine osmolality/specific gravity, calcium, potassium
Hyponatremia	SIADH, adrenal insufficiency, hypothyroidism, hypovolemic states, hypovolemia, medications	Serum Osm, urine Osm, urine Na, glucose, AM cortisol/ACTH if concerning, TSH/free T ₄
Hypokalemic hypertension	Primary aldosteronism, Cushing syndrome, apparent mineralocorticoid excess, renal artery stenosis/diuretics	Aldosterone-renin ratio, bicarbonate, medication review, cortisol screening if Cushing features
Amenorrhea/infertility	Pregnancy, PCOS, hyperprolactinemia, thyroid disease, ovarian insufficiency, hypothalamic amenorrhea, pituitary mass	Pregnancy test, TSH, prolactin, FSH/LH/estradiol, androgen profile if hyperandrogenic
Incidental adrenal mass	Nonfunctioning adenoma, cortisol excess, pheochromocytoma, primary aldosteronism if hypertensive, metastasis/adrenocortical carcinoma	Hormonal screen based on phenotype; imaging density/washout/size; evaluate malignancy history
Incidental thyroid nodule	Benign nodule, autonomous nodule, thyroid cancer, thyroiditis-associated pseudonodule	TSH, ultrasound risk pattern, FNA by size/risk; scan if TSH low

12. Common pitfalls

Pitfall 1: Treating endocrine numbers without the clinical state. A sodium of 118 mEq/L with seizures is managed differently from chronic asymptomatic hyponatremia. A glucose of 500 mg/dL in DKA is different from 500 mg/dL in HHS. A suppressed TSH in a

hospitalized patient can reflect nonthyroidal illness, but fever, delirium, atrial fibrillation, and heart failure may indicate thyroid storm.

Pitfall 2: Forgetting medications. Steroids can cause Cushingoid features and secondary adrenal insufficiency. Amiodarone can cause hypo- or hyperthyroidism. Lithium can cause hypothyroidism, nephrogenic DI, and hyperparathyroidism. Dopamine antagonists can increase prolactin. SGLT2 inhibitors can predispose to euglycemic DKA. Biotin can interfere with some immunoassays.

Pitfall 3: Missing central disease because the tropic hormone is normal. A normal TSH is not reassuring if free T4 is low in a patient with pituitary disease. A normal ACTH may be inappropriate when cortisol is low. A low or normal LH/FSH is abnormal when sex steroids are low.

Pitfall 4: Correcting slowly developing electrolyte disorders too quickly. Chronic hyponatremia develops brain osmolyte adaptation, so rapid correction can cause osmotic demyelination. Hyponatremia and hyperosmolality also require careful correction to avoid cerebral edema or osmotic injury.

Pitfall 5: Starting beta blockade first in pheochromocytoma. Beta blockade without prior alpha blockade can leave alpha-mediated vasoconstriction unopposed and precipitate severe hypertension.

13. Initial endocrine workups by clinical problem

Problem	Initial labs/tests	Interpretive branch points
Suspected diabetes or prediabetes	A1c, fasting plasma glucose; OGTT when needed; symptoms/random glucose in crisis	Confirm abnormal result unless unequivocal hyperglycemia; consider A1c unreliability.
Suspected DKA/HHS	Glucose, BMP, anion gap, beta-hydroxybutyrate/ketones, VBG/ABG, serum osmolality, urinalysis, infection/cardiac trigger evaluation	DKA = ketones + acidosis + gap; HHS = severe hyperosmolality with minimal ketosis/acidosis.
Suspected hyperthyroidism	TSH, free T4, total/free T3 if T3 toxicosis suspected; TRAb/TSI; radioactive iodine uptake if etiology unclear and not pregnant	Low uptake suggests thyroiditis or exogenous hormone; diffuse high uptake suggests Graves; patchy/nodular uptake suggests toxic nodular disease.
Suspected hypothyroidism	TSH and free T4; TPO antibodies if autoimmune context useful	High TSH = primary; low free T4 with low/normal TSH = central.
Suspected Cushing syndrome	One or more validated screening tests: 1-mg overnight dexamethasone suppression, late-night salivary cortisol, or 24-h urine free cortisol	Confirm hypercortisolism before localization; ACTH separates ACTH-dependent from ACTH-independent.
Suspected adrenal insufficiency	8 AM cortisol, ACTH, electrolytes; cosyntropin stimulation when stable	If unstable, draw labs if possible but give hydrocortisone immediately. Hyperkalemia points to primary AI.
Suspected primary aldosteronism	Plasma aldosterone-renin ratio, potassium, bicarbonate; medication preparation if possible	High aldosterone with suppressed renin suggests autonomous aldosterone; confirm and lateralize if surgery considered.
Suspected pheochromocytoma	Plasma free metanephrines or 24-h urine fractionated metanephrines; CT/MRI after biochemical evidence	Measure under appropriate conditions; localize tumor after biochemical diagnosis.
Pituitary mass symptoms	MRI pituitary, visual fields, prolactin, IGF-1, AM cortisol/ACTH, TSH/free T4, LH/FSH/sex steroids	Assess both hypersecretion and hypopituitarism; visual compromise increases urgency.
Hypercalcemia	Repeat calcium with albumin or ionized calcium, PTH, creatinine, phosphate, vitamin D; malignancy workup if PTH suppressed	PTH-mediated vs non-PTH-mediated is the major first split.

14. Integrated practice cases

Case 1. A 43-year-old woman has weight loss, tremor, palpitations, heat intolerance, diffuse goiter, and eye irritation.

Interpretation: The syndrome is thyrotoxicosis, and eye symptoms plus diffuse goiter point toward Graves disease. Initial labs should include TSH and free T4; T3 is useful if TSH is suppressed with normal free T4. Beta blockade controls adrenergic symptoms. Etiology can be supported by TRAb/TSI or diffuse uptake if imaging is needed.

Case 2. A 59-year-old man with type 2 diabetes presents with glucose 870 mg/dL, serum osmolality 340 mOsm/kg, confusion, pH 7.36, trace ketones, and profound dehydration.

Interpretation: This is HHS. The immediate threat is hyperosmolality and volume depletion. Start isotonic fluids, monitor potassium, add insulin after initial fluid and potassium assessment, and search for infection, MI, stroke, or medication trigger.

Case 3. A patient on chronic prednisone stops it abruptly and develops vomiting, abdominal pain, hypotension, and hyponatremia during pneumonia.

Interpretation: This is adrenal crisis from secondary adrenal insufficiency due to HPA suppression. Treat with stress-dose hydrocortisone and IV fluids immediately. Hyperkalemia may be absent because aldosterone is usually preserved in secondary disease.

Case 4. A 36-year-old patient has resistant hypertension and potassium 2.9 mEq/L with metabolic alkalosis.

Interpretation: Primary aldosteronism is a leading diagnosis. Screen with aldosterone-renin ratio. Hypokalemia supports the diagnosis but is not required. If confirmed, adrenal imaging and adrenal venous sampling may help distinguish unilateral adenoma from bilateral hyperplasia.

Case 5. A 29-year-old woman has amenorrhea, galactorrhea, headaches, and bitemporal visual field loss.

Interpretation: Hyperprolactinemia from a macroprolactinoma is likely, but pregnancy, hypothyroidism, and medication causes should be checked. Visual field loss implies mass effect and pituitary MRI is needed. Dopamine agonist therapy is first-line for most prolactinomas.

Case 6. A hospitalized older patient has sodium 121 mEq/L, low serum osmolality, urine osmolality 520 mOsm/kg, urine sodium 58 mEq/L, no edema, normal renal function, and pneumonia.

Interpretation: This is consistent with SIADH after excluding adrenal and thyroid disease. Treat the pneumonia, restrict free water if stable, and use hypertonic saline if severe neurologic symptoms occur. Monitor sodium correction carefully.

Case 7. A patient develops large-volume dilute urine after pituitary surgery and serum sodium rises to 151 mEq/L.

Interpretation: Central diabetes insipidus is likely. Urine osmolality will be inappropriately low for hyponatremia and should rise after desmopressin. Replace free water carefully and treat with desmopressin when indicated.

Case 8. A 67-year-old with constipation, confusion, dehydration, calcium 13.8 mg/dL, suppressed PTH, and weight loss.

Interpretation: Non-PTH-mediated hypercalcemia is likely, and malignancy must be considered. Initial treatment includes volume repletion and temporizing therapy for severe symptoms while evaluating PTHrP, vitamin D-mediated causes, and malignancy.

15. Graph-RAG extraction map

The following map is written so that a graph pipeline can convert endocrine content into nodes, edges, and retrieval anchors. Nodes should preserve numeric thresholds, axis relationships, emergency sequences, and cross-system complications.

Node	Class/subclass	Key attributes to store	Important edges
Endocrine axis	Framework / physiology	Hypothalamus, pituitary, target gland, feedback; tropic vs target hormones	localizes -> primary/secondary/tertiary disease; explains -> lab pair interpretation
Diabetes mellitus	Metabolic / pancreatic endocrine	Diagnostic thresholds, type 1 vs type 2, complications, medication classes	causes -> microvascular/macrovacular disease; treated_by -> lifestyle/metformin/GLP-1/SGLT2/insulin
DKA	Endocrine emergency	Insulin deficiency, ketogenesis, anion gap acidosis, total body K depletion	triggered_by -> infection/insulin omission/MI; treated_by -> fluids/K/insulin
HHS	Endocrine emergency	Severe hyperglycemia, hyperosmolality, dehydration, minimal ketosis	differentiated_from -> DKA; treated_by -> fluids/electrolytes/insulin
Hyperthyroidism	Thyroid excess	Low TSH, high T4/T3, Graves/toxic nodules/thyroiditis/exogenous	causes -> AF/weight loss/tremor; differentiated_by -> uptake pattern
Hypothyroidism	Thyroid deficiency	High TSH primary; low/normal TSH central; symptoms; myxedema coma	treated_by -> levothyroxine; causes -> hyperlipidemia/bradycardia/hyponatremia
Cushing syndrome	Adrenal/pituitary excess	Hypercortisolism, exogenous vs endogenous, ACTH-dependent vs independent	causes -> diabetes/HTN/osteoporosis/infections; localized_by -> ACTH
Adrenal insufficiency	Adrenal/pituitary deficiency	Primary vs secondary, cortisol, ACTH, aldosterone, renin, crisis features	treated_by -> hydrocortisone; primary_causes -> hyperkalemia/hyperpigmentation
Primary aldosteronism	Mineralocorticoid excess	HTN, low renin, high aldosterone, hypokalemia, metabolic alkalosis	causes -> resistant HTN; treated_by -> adrenalectomy or mineralocorticoid receptor antagonist
Pheochromocytoma	Catecholamine tumor	Headache, sweating, palpitations, metanephrines, alpha blockade before beta blockade	causes -> paroxysmal HTN; associated_with -> MEN2/NF1/VHL/SDHx
Pituitary adenoma	Pituitary mass	Micro vs macro, hypersecretion, hypopituitarism, visual field loss	causes -> prolactin excess/acromegaly/Cushing disease; compresses -> optic chiasm
DI	Water-balance disorder	Central vs nephrogenic, dilute urine, hyponatremia, desmopressin response	differentiated_from -> primary polydipsia; treated_by -> DDAVP or thiazide/low solute depending type
SIADH	Water-balance disorder	Hypotonic euvoletic hyponatremia, concentrated urine, urine sodium, no edema	caused_by -> CNS/pulmonary/malignancy/drugs; treated_by -> fluid restriction/hypertonic saline if severe
Hypercalcemia	Calcium disorder	PTH-mediated vs non-PTH-mediated, symptoms, ECG, kidney effects	caused_by -> HPT/malignancy/vitamin D/granulomatous; treated_by -> fluids/calcitonin/antiresorptive

16. High-yield quick tables

Question	Best first mental move
Is the problem hormone excess or deficiency?	Compare symptoms with target-organ effects and pair target hormone with tropic hormone.
Is a gland autonomous?	Look for suppressed upstream signal: low TSH in primary hyperthyroidism, low renin in primary aldosteronism, low ACTH in adrenal Cushing.
Is it pituitary?	Look for inappropriate tropic hormone, multiple pituitary axis problems, visual symptoms, headache, or mass on MRI.
Is it emergency physiology?	Treat shock, arrhythmia, altered mental status, severe dysnatremia, ketoacidosis, and hyperosmolality before elegant localization.
Is the lab artifact-prone?	Consider binding proteins, albumin, biotin, RBC lifespan, renal clearance, medications, acute illness, pregnancy, and timing.

17. Practice questions

1. A patient has low free T4 with a TSH in the lower half of the reference range. What endocrine localization should you consider?

Answer: Central hypothyroidism. Inappropriately normal or low TSH is abnormal when free T4 is low.

2. A patient with severe DKA has potassium 5.6 mEq/L before treatment. Why is potassium still a major danger?

Answer: Total body potassium is depleted from osmotic diuresis and vomiting; insulin and correction of acidosis can shift K intracellularly and cause rapid hypokalemia.

3. What is the first split in hypercalcemia?

Answer: Measure PTH to separate PTH-mediated from non-PTH-mediated hypercalcemia.

4. Why does SIADH usually not cause edema?

Answer: Water retention produces mild volume expansion, which triggers natriuresis and limits sustained extracellular volume expansion.

5. Why can chronic steroid use cause adrenal crisis after abrupt cessation?

Answer: Exogenous glucocorticoids suppress CRH/ACTH and adrenal cortisol production; stress unmasks inability to mount cortisol response.

6. Why should beta blockers not be started before alpha blockade in pheochromocytoma?

Answer: Unopposed alpha stimulation can cause severe vasoconstriction and hypertensive crisis.

7. What test is preferred for screening acromegaly?

Answer: IGF-1 because GH secretion is pulsatile; oral glucose suppression testing helps confirm when needed.

8. What diabetes diagnosis threshold uses symptoms?

Answer: Random plasma glucose ≥ 200 mg/dL with classic symptoms can diagnose diabetes in the right setting.

9. Why might A1c be misleading after transfusion or hemolysis?

Answer: A1c depends on RBC lifespan and glycation time; altered RBC turnover distorts the average.

10. What endocrine cause of hypertension classically produces hypokalemic metabolic alkalosis?

Answer: Primary aldosteronism, although potassium can be normal.

11. What thyroid test pattern suggests Graves disease?

Answer: Low TSH, elevated free T4/T3, positive TSH receptor antibodies, and diffuse radioactive iodine uptake when imaging is used.

12. What should be administered empirically in myxedema coma while starting thyroid hormone?

Answer: Stress-dose glucocorticoids until adrenal insufficiency is excluded.

18. Source and guideline cross-check notes

The endocrine source material for this textbook section included Medvale-uploaded sections on diabetes mellitus, thyroid disease, adrenal disease, and pituitary/DI/SIADH disorders. Current clinical cross-checking used the American Diabetes Association Standards of Care in Diabetes, American Thyroid Association professional guideline resources, and Endocrine

Society endocrine/adrenal/pituitary guideline resources. The purpose of this chapter is educational integration rather than a substitute for local protocols, specialty consultation, or full guideline review.

Selected references

- American Diabetes Association Professional Practice Committee. Standards of Care in Diabetes - current annual guideline resource.
- American Thyroid Association. Professional guidelines and statements for thyroid disease, nodules, and thyroid cancer.
- Endocrine Society. Clinical practice guideline resources for adrenal, pituitary, diabetes, and endocrine disorders.
- Uploaded Medvale source sections: diabetes mellitus; thyroid gland disease; adrenal gland disease; pituitary gland disease, diabetes insipidus, and SIADH.